

Acoustics

The branch of physics that deals with process of generation, reception and propagation of sound is called acoustics. This branch in fact covers many fields and is closely related to various branches of engineering. Some of the important fields of acoustics are:

- i) architectural acoustics dealing with the design and construction of buildings, operas, music halls, recording rooms in radio and television broadcasting stations, auditorium, etc.
- ii) electro-acoustic relating to the production and recording of sound (microphones, amplifiers, loudspeakers, etc).
- iii) design of acoustical instruments and
- iv) musical acoustics dealing with design of musical instruments.

$$p = \frac{2\pi av^2 \rho}{\lambda} \cos \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots(3)$$

The maximum excess of pressure

$$p_{\max} = \frac{2\pi av^2 \rho}{\lambda} \dots\dots\dots(4)$$

and $p = p_{\max} \cos \frac{2\pi}{\lambda} (vt - x) \dots\dots\dots(5)$

again $p_{\max} = 2\pi a \rho v \left(\frac{v}{\lambda}\right)$

$\therefore p_{\max} = 2\pi a \rho v n \dots\dots\dots(6)$

From equations (1) and (6)

$$I = 2\pi^2 \rho n^2 a^2 v$$

$$I = \frac{(2\pi \rho n a v)^2}{2\rho v}$$

$$I = \frac{p_{\max}^2}{2\rho v} \dots\dots\dots(7)$$

The root mean square value of pressure, $p_{rms} = \frac{p_{\max}}{\sqrt{2}}$

Therefore, $I = \frac{p_{\max}^2}{2\rho v} = \frac{(p_{\max} \cdot \sqrt{2})^2}{\rho v} = \frac{p_{rms}^2}{\rho v} \dots\dots\dots(8)$

Equation (7) or (8) shows that the intensity of sound varies directly as the square of the excess pressure. Therefore, in acoustic it is important to measure the excess of pressure to obtain the value of intensity of sound. For ordinary conversation $p_{\max} = 0.1$ newton/m² and the sound output per square metre is 1.13×10^{-5} watt. Human ear is an extremely sensitive organ and can detect intensities as low as 10^{-12} watt/m².

Measurement of Intensity of Sound----Decibel and Phon

The intensity of sound is the quantity of sound energy that flows through a unit area per unit time, the direction of flow being perpendicular to the area. Its unit in SI system is joule/m²-s or watts/m².

The physiological sensation termed loudness depends on the intensity of sound. But the loudness of sound is just an aural sensation and is a physiological phenomenon rather physical phenomenon. The intensity of sound refers to the external or the objective

measurement and the loudness refers to the internal or subjective aspect. Intensity of sound is a definite physical quantity and loudness is merely a degree of sensation. Loudness of sound increases with the intensity of sound according to Weber-Fecher law in physiology. According to this law, the loudness produced is proportional to the logarithm of intensity. This law does not hold good near the lower and the upper limits of audibility. If S stands for loudness when the intensity is I , then

$$S \propto \log I$$

or $S = K \log I$, where K is a constant

$$\text{or } \frac{dS}{dI} = \frac{K}{I}$$

The quantity $\frac{dS}{dI}$ is referred to as the sensitiveness of the ear. As can be seen, the sensitiveness of the ear decreases with the increase in the intensity of sound.

In all cases of practical measurements, it is the relative intensity rather than its absolute value which is more important. Hence intensity of sound is often measured as the ratio to a standard intensity I_0 . The standard value of I_0 is taken as 10^{-12} watt/m², which corresponds to the lowest audible sound at a frequency of 100 Hz. It is referred to as the threshold of hearing.

Bel

Alexander Graham Bell introduced an arbitrary scale of loudness. Whenever the intensity of sound increases by a factor of 10, the increase in intensity is said to be 1 bel. Therefore, the dynamic range of audibility of the human ear is 12 bels or 120 decibels. When the intensity increases by a factor $10^{0.1}$, the increase in intensity is 0.1 bel or 1 decibel.

Suppose the loudness is S for intensity I and S_0 for intensity I_0 , then

$$S = K \log_{10} I$$

$$S_0 = K \log_{10} I_0$$

✓ The intensity level L is the difference in loudness.

∴

$$L = S - S_0$$

$$L = K \log_{10} I - K \log_{10} I_0$$

$$L = K \log_{10} \left(\frac{I}{I_0} \right)$$

When the value of $K = 1$,

$$L = \log_{10} \left(\frac{I}{I_0} \right)$$

Here the unit of L is bel, If L is measured in decibels,

$$L = 10 \log_{10} \left(\frac{I}{I_0} \right) \text{ decibels}$$

Suppose the intensity level changes by 1 decibel, then

$$1 = 10 \log_{10} \left(\frac{I}{I_0} \right)$$

Or $\frac{I}{I_0} = 1.26$

It means that the intensity level alters by 1 decibel when the intensity of sound changes by 26%. The lowest change in intensity level that can be detected by the human ear is 1 decibel.

Phon

The intensity levels of different sounds refer to the loudness in decibels with the assumption that the threshold of audibility is the same, irrespective of the pitch (or frequency) of the sound. However, the sensitivity of the ear and the threshold of audibility vary over wide ranges of frequency and intensity. Hence the intensity level will be different at different frequencies even the same value of I/I_0 . Therefore, for measuring the intensity level, a different unit called the phon is used. The measure of loudness in phons of any sound is equal to the intensity level in decibels of an equally loud pure tone of frequency 1000 Hz. Thus, the phone scale and the decibel scale agree for frequency of 1000 Hz but two values differ at other frequencies. For example, suppose a note of frequency 3000 Hz and the intensity level 70 decibels gives the same loudness as a standard source of frequency 1000 Hz at the intensity level 67 decibels. The intensity level of the note of frequency 3000 Hz is 67 phons.

Reverberation:

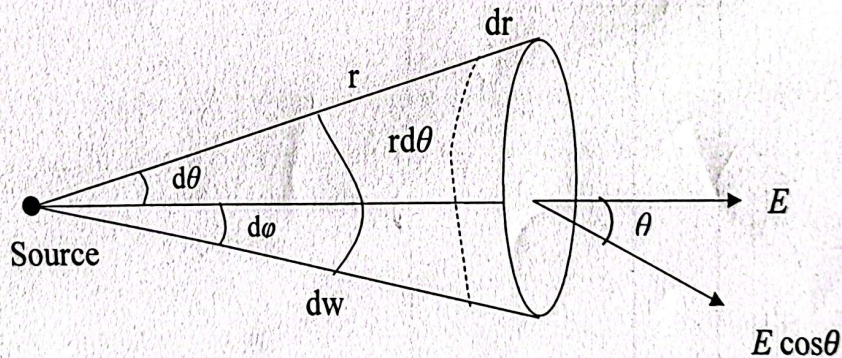
It is observed that for a listener in a room or an auditorium, whenever a sound pulse is produced, he receives directly compressional sound waves from the source, as well as reflected sound waves from the hard walls, ceiling and other materials present in the room. The waves received by the observer are: i) direct waves and ii) reflected waves due to multiple reflections at the various surfaces. Therefore, the listener will receive the combined effect of these two sets of waves. There is also a time gap between the direct wave received by the observer and the waves received by successive reflection. Due to this, the sound persists for sometime even after the source has stopped. This persistence of sound is called reverberation.

The time gap between the initial direct note and the reflected note is called reverberation time. The reverberation time will depend on the size of auditorium, the nature of the reflecting material on the wall and ceiling and the area of the reflecting surfaces. An auditorium is more suitable to keep the reverberation time to be more small.

Sabine's Reverberation Formula: (book from L P Sharma & H C Saxena)

Sabine developed the reverberation formula to express the rise and fall of sound in an auditorium after making some assumptions:

- the distribution of sound energy and intensity is uniform inside enclosure.
- the dissipation of energy in air in the enclosure is absolutely negligible. Only lost of energy is only due to the absorption of the material of the walls, ceiling and other materials.
- the rate of emission of sound energy is constant.



If E represents the energy density, then the energy in an element of volume dV is EdV . Let two wave fronts are placed at distance r and $r+dr$ from the source of sound. So, the area of this element which is enclosed by these two wave fronts is $rd\theta.dr$.

If the whole figure is rotated about the normal through a small angle $d\phi$, the elementary volume produced by the surface element $rd\theta.dr$ is given by $dV = rd\theta.dr.r\sin\theta d\phi$.

Energy in the element dV is obtained by $Er^2 \sin\theta dr d\theta d\phi$.

The solid angle, dw subtended by the area dS is $\frac{dS \cos\theta}{r^2}$ at that element. Since the energy is transmitted uniformly in all direction, the energy passing through unit solid angle at any instant is given

$$\frac{d\omega}{4\pi} EdV = \frac{dS \cos\theta}{4\pi} \times Er^2 \sin\theta d\theta dr d\phi$$

$$= \frac{EdS}{2 \times 4\pi} dr \sin 2\theta d\theta d\phi$$

The total energy received by the element dS in one second is from a hemisphere of radius v , where v is the velocity of sound.

Energy received by dS in one second is,

$$\begin{aligned} &= \frac{EdS}{2 \times 4\pi} \int_0^v dr \int_0^{\frac{\pi}{2}} \sin 2\theta \int_0^{2\pi} d\phi \\ &= \frac{EdS}{2 \times 4\pi} [r]_0^v \left[\frac{-\cos 2\theta}{2} \right]_0^{\frac{\pi}{2}} [\phi]_0^{2\pi} \\ &= \frac{EdS}{2 \times 4\pi} v \left[-\frac{1}{2} (\cos \pi - \cos 0) \right] 2\pi \\ &= \frac{Ev dS}{4} \dots \dots \dots (1) \end{aligned}$$

If a is the absorption co-efficient of the materials of the walls and ceiling. The energy in the whole enclosure per second

$$= \int \frac{EvadS}{4} = \frac{EvaS}{4}$$

Let p be the rate of emission of energy from the source and V be the volume of the enclosure, the energy in the enclosure at any instant is EV . Hence, the rate of growth of energy is

$$\frac{d}{dt}(EV) = \frac{VdE}{dt}$$

Now,

Rate of emission of energy by the source = Rate of growth of energy in space + Rate of absorption of energy by the walls

$$\begin{aligned} \therefore \rho &= \frac{VdE}{dt} + \frac{EvaS}{4} \\ \frac{dE}{dt} + \frac{EvaS}{4V} &= \frac{\rho}{V} \end{aligned}$$

Let $\frac{vaS}{4V} = b$

Then we have $\frac{dE}{dt} + bE = \frac{4b\rho}{vaS}$

Multiplying both sides by e^{bt} we have

$$e^{bt} \frac{dE}{dt} + Eb e^{bt} = \frac{4b\rho}{vaS} e^{bt}$$

$$\frac{d}{dt}(E e^{bt}) = \frac{4b\rho}{vaS} e^{bt}$$

On integrating we have

$$E e^{bt} = \frac{4\rho}{vaS} e^{bt} + K$$

Where K is a constant of integration,

At $t = 0, E = 0$

$$K = -\frac{4\rho}{vaS}$$

Therefore, $Ee^{bt} = \frac{4\rho}{vaS} (e^{bt} - 1)$

$$\therefore E = \frac{4\rho}{vaS} (1 - e^{-bt})$$

Putting $\frac{4b\rho}{vaS} = E_s$

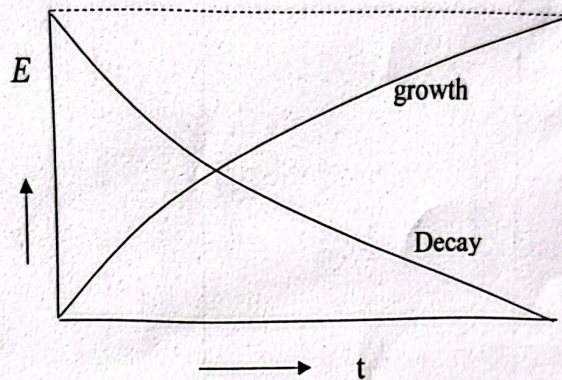
$$\therefore E = E_s(1 - e^{-bt}) \dots\dots\dots(2)$$

The growth and decay of energy is exponential. After a longtime i.e., $t \cong \infty$.

$$E = E_s = E_{\max} \dots\dots\dots(3)$$

If the source ceases to emit energy, the decay of energy per unit volume is given

$$E = E_s e^{-bt} \dots\dots\dots(4)$$



The reverberation time which is defined to be the time during which the steady energy density drops down to one millionth of its value is given by

$$e^{-bt} = \frac{E}{E_s} = 10^{-6} \text{ and } t = T$$

$$-\frac{vaS}{4V} T = -6 \log_e 10$$

$$\frac{vaS}{4V} T = 2.303 \times 6$$

$$T = \frac{4 \times 2.303 \times 6 \times V}{vaS}$$

$$= \frac{4 \times 2.303 \times 6 \times V}{1120 \times a \times S}$$

$$= \frac{0.05V}{aS}$$

In M.K.S unit

$$v = 350 \text{ m/sec}$$

$$b = \frac{vaS}{4V}$$

T = reverberation time

$$v = 1120 \text{ ft/sec}$$

$$T = \frac{4 \times 2.303 \times 6 \times V}{350 \times aS}$$

$$= \frac{0.158}{aS}$$

Example: The volume of a room is 600 m^3 , the wall area of the room is 220 m^2 , the floor is 120 m^2 , and the ceiling area is 120 m^2 . The average sound absorption co-efficient, i) for the walls is 0.03, ii) for the ceiling is 0.80 and iii) for the floor is 0.06. Calculate the average sound absorption co-efficient and the reverberation time.

The average sound absorption co-efficient

$$a = \frac{\sum a_i s_i}{\sum s_i}$$

$$= \frac{a_1 s_1 + a_2 s_2 + a_3 s_3}{s_1 + s_2 + s_3}$$

Here,

$$s_1 = 220 \text{ m}^2$$

$$a_1 = 0.03$$

$$s_2 = 120 \text{ m}^2$$

$$a_2 = 0.80$$

$$s_3 = 120 \text{ m}^2$$

$$a_3 = 0.06$$

$$\therefore a = \frac{0.03 \times 220 + 0.80 \times 120 + 0.06 \times 120}{220 + 120 + 120}$$

$$= \frac{109.8}{460} = 0.2389$$

$$= 0.24$$

We know the reverberation time

$$T = \frac{0.158 V}{aS}$$

Here, $a = 0.24$

$$S = \sum s_i = s_1 + s_2 + s_3 = (220 + 120 + 120) \text{ m}^2$$

$$= 460 \text{ m}^2$$

$$\therefore T = \frac{0.158 V}{aS}$$

$$= \frac{0.158 \times 600}{0.24 \times 460} = 0.85869$$

$$= 0.86 \text{ sec}$$

Requisites for good acoustics

The reverberation time can be reduced by the following method.
Book $\rightarrow$ Physics for Engineering by Prof. Gias Uddin, page 314.